R and Stan Source Codes

arXiv: https://arxiv.org/abs/2604.09259

This document describes the structure and content of the supplementary R source codes provided with article entitled *"Exact Bayesian Planning for Simple Step-Stress Accelerated Life Testing with Competing Risks"*

General Information

All computations were carried out in R, using parallel computing via the `foreach` and `doParallel` packages, with posterior inference performed using the No-U-Turn Sampler (NUTS) implemented in Stan via the `cmdstanr` package. The full two-variable optimal design computation under a single prior specification required approximately 8 hours on the High-Performance Computing cluster (Comet) at Newcastle University using 48 cores with 16 parallel workers, each handling 3 parallel chains.

In addition to base R, the analyses use `ggplot2`, `cowplot` and `lattice` for graphical outputs and `dplyr` for data handling. The supplementary material is provided as a compressed archive containing this README document and the following files.

Dependencies

R packages

- `cmdstanr` — R interface to CmdStan for running Stan models
- `parallel`, `doParallel`, `foreach` — parallel computing
- `ggplot2`, `cowplot`, `lattice` — plotting
- `dplyr` — data manipulation

Stan

CmdStan (version 2.33.1 or later) is required. Install `cmdstanr` and CmdStan following the instructions at: https://mc-stan.org/cmdstanr/articles/cmdstanr.html

File Descriptions

File	Description
`bernoulli_han_ssalt_gamma_repara.stan`	Stan model for posterior inference
`all_functions.R`	All R helper functions
`bootstrap_mle_se_for_Han_with_while_loop.R`	MLE fitting and prior elicitation
`convergence_diago_han_ssalt_gamma.R`	Convergence diagnostics
`design_han_ssalt_gamma_repaar_Prior_I.R`	Main script: optimal design

Stan model: The Stan model declares the reparametrised quantities $\boldsymbol{\varphi}_j = (t^q_{0,j}, -b_j, \beta_j)$, recovers the original model parameters (a_j, b_j, β_j), computes the p-th quantile $t_p(x_0)$ and $\log(t_p(x_0))$ at use stress, and evaluates the likelihood under the cumulative exposure model.

Helper functions: The file `all_functions.R` contains all supporting R functions including the SSALT data simulation function `simSSAT_CR`, likelihood evaluation, MLE computation, and criterion function evaluation routines used throughout the main script.

MLE fitting and prior elicitation: The script `bootstrap_mle_se_for_Han_with_while_loop.R` fits the step-stress competing risks Weibull model to the preliminary dataset by maximum likelihood, performs a parametric bootstrap with 1000 replicates to characterise the sampling variability of the MLEs of $\boldsymbol{\varphi}_j$, and computes the prior hyperparameters via method-of-moments. This script reproduces Table 2 and Figure 2 of the paper.

Convergence diagnostics: The script `convergence_diago_han_ssalt_gamma.R` runs the NUTS sampler for a representative fixed design point ($\tau = 5$, $s_1 = 1/320\ \mathrm{K}^{-1}$) and produces the posterior summary statistics, trace plots, density plots, and autocorrelation function plots reported in Section 4 and Table 3 of the paper.

Main design script: The main R script performs the Monte Carlo simulation loop over the τ grid, calls the Stan model at each grid point for $B = 1000$ replications, computes the preposterior criterion values $C_1(\tau)$ and $C_2(\tau)$, applies the Gaussian kernel smoother, and locates the optimal design τ^0. It reproduces the sensitivity analyses SA1–SA4 reported in Section 5 of the paper.

Data

The case study uses the step-stress ALT dataset from:

> Han, D. and Kundu, D. (2015). Inference for a step-stress model with competing risks for failure from the generalized exponential distribution under Type-I censoring. *IEEE Transactions on Reliability*, 64(1), 31–43.
> DOI: 10.1109/TR.2014.2336392

Reproducibility

All R scripts are written to reproduce the results reported in the manuscript. Set the global seed to `2026` to reproduce the exact numerical results. Since the numerical experiments rely on random number generation, minor variation across runs is expected, but the overall patterns, optimal designs, and conclusions remain unchanged. The codes are provided to ensure transparency and to allow readers to reproduce and verify the reported findings.

Contact

For any questions regarding the codes, readers may contact the corresponding author.

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